

Coal Mine Safety and Health Act 1999
Electrical Engineering Manager Legislation Exam

12 Meaning of *competence*

Competence for a task at a coal mine is the demonstrated skill and knowledge required to carry out the task to a standard necessary for the safety and health of persons.

15 Meaning of *accident*

An *accident* at a coal mine is an event, or a series of events, at the coal mine causing injury to a person.

16 Meaning of *serious accident*

A *serious accident* at a coal mine is an accident at a coal mine that causes—

- (a) the death of a person; or
- (b) a person to be admitted to a hospital as an in-patient for treatment for the injury.

17 Meaning of *high potential incident*

A *high potential incident* at a coal mine is an event, or a series of events, that causes or has the potential to cause a significant adverse effect on the safety or health of a person.

26 Meaning of *supervisor*

A *supervisor* at a coal mine is a person appointed under [section 56](#) to—

- (a) implement and monitor the coal mine's safety and health management system; and
- (b) give directions to other coal mine workers at the coal mine in accordance with the safety and health management system.

30 How is an acceptable level of risk achieved

- (1) To achieve an acceptable level of risk, this Act requires that management and operating systems must be put in place for each coal mine.
- (2) This Act provides that the systems must incorporate risk management elements and practices appropriate for each coal mine to—
 - (a) identify, analyse, and assess risk; and
 - (b) avoid or remove unacceptable risk; and
 - (c) monitor levels of risk and the adverse consequences of retained residual risk; and
 - (d) provide for critical controls; and
 - (e) investigate and analyse the causes of serious accidents and high potential incidents with a view to preventing their recurrence; and
 - (f) review the effectiveness of risk control measures, and take appropriate corrective and preventive action; and
 - (g) mitigate the potential adverse effects arising from residual risk.
- (3) Also, the way an acceptable level of risk of injury or illness may be achieved may be prescribed under a regulation.

39A Additional obligations

(1) This section applies to each of the following persons—

- (a) a coal mine worker at a coal mine;
- (b) another person at a coal mine;
- (c) an ROC worker for a coal mine.

(2) The person has the following additional obligations—

- (a) to work or carry out the person's activities in a way that does not expose the person or someone else to an unacceptable level of risk;
- (b) to ensure, to the extent of the responsibilities and duties allocated to the person, that the work and activities under the person's control, supervision, or leadership is conducted in a way that does not expose the person or someone else to an unacceptable level of risk;
- (c) to the extent of the person's involvement—to participate in and conform to the risk management practices of the mine;
- (d) to comply with instructions given for safety and health of persons by the coal mine operator or site senior executive for the mine or a supervisor at the mine;
- (e) to work at or for the coal mine only if the person is in a fit condition to carry out the work without affecting the safety and health of others;
- (f) not to do anything wilfully or recklessly that might adversely affect the safety and health of someone else at the mine.

55 Management structure for safe operations at coal mines

(1) The site senior executive for a coal mine must—

- (a) develop and maintain a management structure for the coal mine in a way that allows development and implementation of the safety and health management system; and
- (b) document the management structure.

Maximum penalty—40 penalty units.

(2) The document must state—

- (a) the responsibilities of the site senior executive; and
- (b) the responsibilities and competencies required for senior positions in the structure; and
- (c) the names of the persons holding the senior positions and their competencies; and
- (d) the name of the person who is responsible for establishing and implementing a system for managing contractors at the coal mine; and
- (e) the competencies required, and the responsibilities, for each other supervisory position at the mine.

Maximum penalty—40 penalty units.

(3) For subsection (2)(b), an inspector may by notice given to the site senior executive declare a position to be a senior position.

(4) For each supervisory position mentioned in subsection (2)(e), the site senior executive must also keep a record of the names and competencies of each person authorised to carry out the responsibilities of the position.

200 Site not to be interfered with without permission

- (1) A person must not interfere with a place at a coal mine that is the site of a serious accident or high potential incident of a type prescribed by regulation, without the permission of an inspector.

Maximum penalty—200 penalty units.

- (2) Permission under subsection (1) must not be unreasonably withheld.

- (3) For this division, action taken to save life or prevent further injury at a place is not interference with the place.

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16A Supervisor's responsibilities

A supervisor for a coal mine has the following responsibilities—

- (a) ensuring each mine worker for whom the supervisor has responsibilities is given the supervision and help mentioned in [sections 85A\(2\)](#) and [85B](#);
- (b) observing each coal mine worker for whom the supervisor has responsibilities perform tasks;
- (c) carrying out inspections of workplaces at the coal mine as required under the coal mine's safety and health management system;
- (d) recording reports of, and findings and recommendations resulting from, inspections mentioned in paragraph (c) in the mine record;
- (e) implementing and monitoring hazard controls and risk controls as required under the coal mine's safety and health management system.

17 Qualifications for appointment as electrical engineering manager

A person may appoint a person as an electrical engineering manager only if the person has the recognised competencies for controlling and managing the mine's electrical engineering activities.

18 Duties of electrical engineering manager

The duties of an electrical engineering manager include controlling and managing the following at the mine—

- (a) the design of electrical installations;
- (b) the installation and maintenance of electrical equipment and electrical installations;
- (c) electrical work;
- (d) work carried out close to electrical installations.

19 Duty to give notice of proposed introduction of electricity

Before electricity is introduced at a coal mine for use in coal mining operations, or after electricity is permanently disconnected, the site senior executive for the coal mine must notify an inspector of the proposed introduction, or disconnection.

20 Competencies of persons carrying out work on electrical equipment

(1) Subject to subsection (2), only a person who has the recognised competency for repairing, overhauling and maintaining electrical equipment for explosive atmospheres may carry out the following work at a coal mine—

- (a) electrical work in an ERZ;
- (b) work on electrical equipment used in an ERZ;
- (c) work on extra low voltage systems associated with explosion-protected equipment and installations;
- (d) work on explosion-protected electrical equipment installed at a surface mine;
- (e) electrical work in an area, at a surface mine, that a risk assessment shows has an explosion risk.

(2) Only a person who has the recognised competency for repairing, overhauling and maintaining electrical equipment for mobile plant used in underground mines may carry out work on extra low voltage systems associated with mobile plant for an underground mine.

21 Access to exposed electrical conductors

(1) A coal mine must have a standard operating procedure for safely accessing exposed electrical conductors at the mine.

(2) The procedure must provide for minimising the risk of energising exposed electrical conductors on which work is to be carried out, including, for example, by using—

- (a) personal locks; or
- (b) danger tags; or
- (c) personal locks and danger tags.

(3) The procedure must also provide that before work is carried out on an exposed electrical conductor at above extra low voltage the conductor must be—

- (a) positively isolated from the electricity source; and
- (b) tested for zero potential; and
- (c) if the conductor is a high voltage conductor, earthed.

(4) Subsection (5) applies only to a surface mine and only if—

- (a) it is not practicable to positively isolate the conductor from the electricity source; and
- (b) access to the conductor is required or work in close proximity to the conductor is required to be carried out.

(5) The procedure must also provide that—

- (a) a risk assessment, including consideration of high fault currents, must be carried out before the work is carried out; and
- (b) if the risk assessment shows an unacceptable level of risk in carrying out the work with the conductor energised, before the work is carried out, the conductor must be—
 - (i) positively isolated from the electricity source; and
 - (ii) tested for zero potential; and
 - (iii) if the conductor is a high voltage conductor, earthed.

(6) In this section—

close proximity, to a conductor, means—

- (a) for a person—
 - (i) for a high voltage conductor—within the exclusion zone for the person for the conductor under the [Electrical Safety Regulation 2013](#); or
 - (ii) for a low voltage conductor—within reach of the conductor; or
- (b) for plant being used for work near an overhead power line—within the minimum clearances required under AS/NZS 3007:2013.

22 Locating and dealing with electrical faults

(1) A coal mine must have a standard operating procedure for re-energising an electrical circuit at the mine that has been switched off by a circuit protection device.

(2) A person may use electrical energy, other than electrical energy generated from appropriately designed test instruments, for locating, isolating or clearing only the following electrical faults—

- (a) an overload of an electric motor;
- (b) an earth leakage trip associated with an impedance earth circuit;
- (c) a 30mA earth leakage trip;
- (d) an earth continuity trip.

23 Design, installation and maintenance

The site senior executive for a coal mine must ensure the design, installation and maintenance of electrical equipment and installations at the mine provide for the following—

- (a) reliable circuit interruption, under fault conditions, at all points in the mine's electrical distribution system;
- (b) each earthing system at the mine to be of sufficiently low impedance to ensure reliable operation of all electrical protection systems and devices and adequate protection against indirect contact;
- (c) limiting prospective touch voltage, under fault conditions, to within acceptable limits;
- (d) minimising potential impacts from voltage rise caused by lightning strike, static electricity, voltage surges and other transient voltages to within acceptable limits;
- (e) preventing a person inadvertently contacting live parts of equipment or an installation exceeding extra low voltage.

Example for paragraph (e)—

The design, installation and maintenance may provide for insulation or an enclosure, shield, interlocking device or automatic disconnection device for the equipment or installation.

24 Isolators for equipment driven by electricity

- (1) A coal mine's electrical engineering manager must ensure the mine has a full current isolator for equipment driven by electricity at the mine.
- (2) The electrical engineering manager must ensure—
 - (a) the isolator is—
 - (i) clearly identified as the isolator for the equipment; and
 - (ii) situated in a location that is easily accessible by a person working on the equipment; and
 - (b) the equipment is clearly identified as being supplied with electricity from the isolator.

Example of how the isolator or equipment may be identified for subsection (2)—

by marking the isolator or equipment, or placing a notice near the isolator or equipment

25 Standard for electrical control systems

A coal mine's electrical engineering manager must ensure—

- (a) each electrical control system at the mine operates safely under all operating conditions, including electricity supply instability or failure; and
- (b) if the system suffers a fault or fails, all emergency stopping systems and safety alarms at the mine remain effective.

26 Control circuits

- (1) A coal mine's electrical engineering manager must ensure a control circuit at the mine run externally to an enclosure from which the circuit originates—
 - (a) has a voltage of not more than a nominal voltage of 55V to earth; or
 - (b) is protected by earth leakage current protection of not more than 30mA sensitivity; or
 - (c) minimises the risk of direct contact and limits prospective touch voltage to within acceptable limits.
- (3) An underground mine's electrical engineering manager must ensure a control circuit for the mine is explosion protected, unless the whole circuit is installed and operated in a NERZ.

27 Modification of electrical control systems

A coal mine's safety and health management system must—

- (a) provide for the security and maintenance of the mine's electrical control system software and control circuits; and
- (b) control modification of the software and circuits; and
- (c) provide for records to be kept of any modifications.

28 Electrical protection for power outlets for low voltage electrical equipment

(1) A coal mine's electrical engineering manager must ensure a single-phase power outlet for low voltage electrical equipment at the mine has—

- (a) earth leakage current protection of not more than 30mA sensitivity; or
- (b) a device preventing its use by portable electrical equipment.

(2) Subsection (1) does not apply if the power outlet is the only outlet in an isolated electricity supply.

(3) In this section—

isolated electricity supply means an electricity supply—

- (a) other than the main electricity supply, that is totally insulated from conductive material; or
- (b) that is—
 - (i) supplied with electricity from the main electricity supply; and
 - (ii) totally isolated from the main electricity supply; and
 - (iii) totally insulated from earth.

29 Operating times and tripping current for circuit protection devices

(1) A coal mine's electrical engineering manager must ensure each circuit protection device at the mine has an appropriate operating time and tripping current, having regard to a risk assessment of the operation of the part of the electrical installation the device protects.

(2) For a surface mine, the operating time may allow for a delay in cutting off the electricity supply if—

- (a) the device is installed to protect a person operating equipment; and
- (b) it is appropriate to allow the person to bring the equipment to rest before the electricity supply is cut; and
- (c) the person is immediately given a clear warning of the fault.

30 Changing electrical protection settings

(1) A person must not change an electrical protection setting for a circuit protection device at a coal mine, unless the person is authorised to make the change by the electrical engineering manager for the mine.

(2) In this section—

electrical protection setting, for a circuit protection device, means the current, voltage or operating time at which the device is set to trip when the device detects an electrical fault in the circuit.

31 Unearthed electrical installations

(1) A coal mine's electrical engineering manager must ensure an unearthed electrical installation at the mine has earth fault detection and control.

(2) Subsection (1) does not apply to a single-phase 240V generator of less than 25kW capacity at a surface mine.

32 Drawings of electrical installations

- (1) The site senior executive for a coal mine must ensure up-to-date drawings of the mine's electrical installations are available for use at the mine when electrical equipment and electrical installations are commissioned or modified.
- (2) The drawings must be updated as soon as practicable, but not later than 8 days, after—
 - (a) the commissioning of any electrical equipment or electrical installation at the mine; or
 - (b) any major modification to electrical equipment or an electrical installation at the mine.
- (3) The drawings must be kept in a location that is easily accessible by each coal mine worker authorised by an electrical supervisor or the coal mine's electrical engineering manager to access the drawings.

34 Records

- (1) A coal mine's electrical engineering manager or, if there is no electrical engineering manager for the mine, the site senior executive for the mine, must ensure a record of the following matters about the mine's electrical activities, equipment and installations is kept at the mine—
 - (a) for each electrical installation—
 - (i) its electrical protection settings, fault levels, and cable and conductor sizes and switchgear ratings, including circuit interrupting capacity; and
 - (ii) its commissioning dates and results; and
 - (iii) its monitoring and test dates and results;
 - (b) for each item of electrical equipment—
 - (i) its commissioning dates and results; and
 - (ii) its monitoring and test dates and results;
 - (c) for each item of electrical safety equipment used—
 - (i) its description; and
 - (ii) its test dates and results;
 - (d) the name of each person appointed—
 - (i) to control and manage the mine's electrical engineering activities; or
 - (ii) to carry out electrical work and electrical activities;
 - (e) the name of each person authorised to enter an electrical operating area;
 - (f) the activity each person mentioned in paragraph (e) is authorised to carry out in the operating area.
- (2) In this section—

electrical operating area means an area or enclosure in which electrical equipment or an electrical installation is housed.

electrical safety equipment includes an instrument for testing electrical equipment, a ladder, a safety harness, an insulating tool and insulating gloves.

35 General

- (1) A coal mine's safety and health management system must provide for managing emergencies at the mine.
- (2) The system must provide for the following—
 - (a) identifying, by risk assessment, potential emergency situations;
 - (b) minimising risks associated with potential emergency situations;
 - (c) carrying out aided rescue and self-escape of persons from the mine in an emergency;
 - (d) carrying out emergency exercises, including testing the effectiveness of emergency management procedures and the readiness and fitness of equipment for use in an emergency;
 - (e) auditing and reviewing the emergency exercises;
 - (f) if the mine is a surface mine—involving an open-cut examiner for the mine in—
 - (i) developing and testing the emergency management procedures for activities, including mining activities, in and around the surface excavation at the mine; and
 - (ii) auditing the documentation for the procedures;
 - (g) if the mine is an underground mine—involving an ERZ controller for the mine in—
 - (i) developing and testing the emergency management procedures for explosion risk zones; and
 - (ii) auditing the documentation for the procedures.

37 Fire

- (1) A coal mine's safety and health management system must provide for the following at the mine—
 - (a) fire prevention and control;
 - (b) an effective fire fighting capability;
 - (c) the safety of persons fighting fires;
 - (d) a risk assessment to identify all potential fire hazards at the mine.
- (2) The system must also provide for the following—
 - (a) the availability at the mine, at all times, of equipment that is appropriate and sufficient to extinguish any potential fire identified by the risk assessment;
 - (b) the location of portable fire extinguishers on or near equipment and installations identified as potential fire hazards by the risk assessment;
 - (c) the compatibility, throughout the mine, of all fire fighting equipment.
- (3) The coal mine must have a standard operating procedure for action to be taken when a fire is discovered at the mine.

38 First aid

- (1) A coal mine's safety and health management system must provide for first aid at the mine.
- (2) The system must provide for the following—
 - (a) the supply and placement of first aid supplies and equipment, including supplies and equipment required for emergencies, based on a risk assessment;
 - (b) appropriate first aid training, including refresher training, emphasising the safety of the person giving first aid as well as the safety of the person receiving first aid;
 - (c) the availability of a sufficient number of persons competent in giving first aid.
- (3) For accidents involving electricity, the mine must have a standard operating procedure for the following—
 - (a) releasing a person from an energised low and high voltage conductor;
 - (b) resuscitating a person in an electrical environment;
 - (c) managing a person's flash burn injuries.

68 Modifying plant

- (1) A coal mine must have a standard operating procedure for modifying fixed and mobile plant.
- (2) A coal mine's safety and health management system must provide for the following—
 - (a) recording modifications made to the plant at the mine;
 - (b) updating drawings of the plant held at the mine to include the modifications;
 - (c) assessing and managing risk associated with the modifications.

69 Pre-start warning

- (1) This section applies if starting fixed or mobile plant at a coal mine is likely to cause a hazard to a person near the plant.
- (2) The mine's safety and health management system must provide for the plant to be fitted with a device that sounds a warning before the plant is started.

78 Isolating and tagging procedures

- (1) A coal mine must have a standard operating procedure for the following—
 - (a) controlling the risk of an unplanned release of energy from plant, including positively isolating the energy source;
 - (b) if an electrical or mechanical energy source is positively isolated—testing for zero potential;
 - (c) taking plant out of service;
 - (d) returning plant to service.
- (2) Without limiting subsection (1), the standard operating procedure may provide for the use of danger, isolation, operational, out of service, personal and restriction tags for particular circumstances.
- (3) If the safety or health of a person is directly affected by the operation or non-operation of plant, the procedure must also provide for the person to personally control, by attaching a danger tag or lock to the plant, the plant's change in status from non-operational to operational.
- (4) For subsection (1)(a), the method for positively isolating the energy source for plant provided for in the standard operating procedure must—
 - (a) for plant that is electrical equipment—be a manually initiated operation that—
 - (i) isolates all active power conductors of the plant being isolated from the electricity supply; and
 - (ii) prevents unintended re-energisation, including re-energisation through inadvertent operation of the plant or component failure; and
 - (b) for other plant—be the operation of a manually operated device that—
 - (i) is installed in the energy supply for the plant and, when operated, isolates the plant from its energy source; and
 - (ii) requires a manually initiated operation for the supply of energy to the plant to be resumed; and
- (5) In this section—

isolation device means—

- (a) for plant that is electrical equipment—the device used manually to initiate the operation mentioned in subsection (4)(a); or
- (b) for other plant—the device mentioned in subsection (4)(b).

79 Equipment used for maintaining plant

- (1) A coal mine must have a standard operating procedure for electrical and mechanical equipment used for inspecting, testing and maintaining the safe operation of plant at the mine.
- (2) The procedure must provide for the following—
 - (a) selecting equipment that is fit for its intended purpose;
 - (b) using the equipment safely;
 - (c) maintaining the equipment in accordance with its specification;
 - (d) testing the equipment at appropriate intervals and labelling the equipment to show when the equipment was last tested.

85A Time and resources for carrying out tasks

- (1) The site senior executive for a coal mine must ensure time is allocated, and the mine's resources are distributed, to enable each coal mine worker at the mine to carry out the worker's tasks without creating an unacceptable level of risk.
- (2) Without limiting subsection (1), the site senior executive must ensure the coal mine worker is given the supervision, and help from other competent persons, necessary to achieve an acceptable level of risk.

85B Supervising workers

- (1) Without limiting [section 85A](#)(2), the site senior executive for a coal mine must ensure the coal mine's activities and coal mine workers are supervised to the extent necessary to ensure each worker—
 - (a) is not likely to be exposed to conditions beyond the worker's capabilities; and
 - (b) is not likely to be affected by the conditions in which the worker is working in a way that adversely affects the worker's fitness to perform critical tasks; and
 - (c) has the resources necessary to undertake the work without being exposed to an unacceptable level of risk; and
 - (d) is working within the limits of the worker's fitness and competence; and
 - (e) complies with the worker's safety and health obligations.
- (2) The supervision must include communication, including direct contact, between the supervisor and the coal mine worker at appropriate intervals.

86 Air conditioning units

A coal mine's safety and health management system must provide for maintaining and cleaning air conditioning units used at the mine.

88 Cutting, drilling or excavating near a concealed service

- (1) A coal mine's safety and health management system must provide for controlling risks from cutting or drilling into a building or structure, or excavating ground, at the mine where there may be a concealed service.
- (2) The system must provide for the following to minimise the uncontrolled release of non-electrical or electrical energy from a concealed service—
 - (a) locating the energy source and placing a mark, describing the source, on or near the source;
 - (b) isolating the energy source;
 - (c) testing the energy source and proving the source de-energised.
- (3) The site senior executive for the coal mine must provide sufficient equipment for the matters mentioned in subsection (2)(a) to (c).

112 Live testing electrical equipment

- (1) A surface mine must have a standard operating procedure for live testing electrical equipment at the mine.
- (2) The procedure must—
 - (a) state—
 - (i) what equipment may be live tested; and
 - (ii) what instruments may be used for the testing; and
 - (b) provide for the following—
 - (i) isolating areas where equipment is being live tested or test run;
 - (ii) using personal protective equipment;
 - (iii) using high voltage test equipment and discharging stored energy after the test;
 - (iv) dealing with high prospective fault currents;
 - (v) other precautions to be taken for live testing.

113 Electrical protection for electrical distribution system

- (1) The site senior executive for a surface mine must ensure an earthed electrical distribution system at the mine has earth leakage protection.
- (2) Subsection (1) does not apply if—
 - (a) persons at the mine are protected against indirect contact with the system; or
 - (b) for an overhead power line—
 - (i) mobile plant is not operated close to the line; and
 - (ii) a risk assessment shows earth leakage protection would only minimally reduce the risk.

114 Electrical protection for mobile or transportable equipment

- (1) This section applies to mobile or transportable equipment used at a surface mine and supplied with electricity by a reeling or trailing cable.
- (2) If the equipment does not have a local earthing system, the site senior executive for the mine must ensure the cable has an earth continuity monitoring system that cuts off the electricity supply immediately after an earth continuity fault is detected in the cable or equipment.
- (3) Subsection (2) does not apply if the cable—
 - (a) is a low voltage cable; and
 - (b) has earth leakage current protection of not more than 30mA sensitivity; and
 - (c) has individually screened cores; and
 - (d) has no joining plugs or sockets.
- (4) The site senior executive must ensure the electricity source for the equipment has an earth fault current limitation device that limits prospective touch voltage to within acceptable limits.
- (5) Subsection (4) does not apply if the equipment is low voltage equipment and persons at the mine are otherwise protected adequately against injury from indirect contact with electricity.

127 Plans of surface services

(1) The site senior executive for a surface mine must ensure current plans identifying and showing the location of surface services, including buried services, are kept in a location at the mine that is easily accessible by each coal mine worker at the mine.

(2) In this section—

buried service includes electricity conducted through an underground electrical cable and another service carried in an underground pipe.

surface service includes the following—

- (a) a communication cable;
- (b) compressed air storage and reticulation;
- (c) diesel storage and reticulation;
- (d) an electrical cable;
- (e) an emergency access or road;
- (f) a fire fighting facility;
- (g) a first aid facility;
- (h) gas storage and reticulation;
- (i) rescue equipment;
- (j) a sewerage facility;
- (k) a water storage pipeline or drain.

132 Using and relocating operator-attended dredges

(1) A surface mine must have a standard operating procedure for using and relocating operator-attended dredges at the mine.

(2) The procedure must provide for the following—

- (a) safe access to a floating pontoon;
- (b) restricting access to areas identified in a risk assessment as high risk areas;
- (c) controlling electrical trailing cables connected to the dredge;
- (d) the safety of persons near winch ropes;
- (e) using a bilge pump, including, in particular, ways to minimise the risk of reverse siphoning;
- (f) checking and recording the dredge's freeboard;
- (g) maintaining and inspecting the dredge's hulls and parts essential for flotation;
- (h) controlling risk from dredging the face;
- (i) monitoring climatic conditions and securing the dredge during adverse climatic conditions.

135 Using plant near electricity

- (1) A surface mine must have a standard operating procedure for using fixed and mobile plant near electrical cables, overhead electrical conductors, or electrical equipment.
- (2) The procedure must provide for the following—
 - (a) recording power line heights at each location where a vehicle may pass;
 - (b) keeping a register of the height of vehicles and mobile plant used on site;
 - (c) fixing standard travelling routes at the mine;
 - (d) travelling other than on a standard travelling route;
 - (e) inspecting, and reporting on, travelling routes for plant not included in the register mentioned in paragraph (b);
 - (f) preventing incidents involving electrical contact, including, for example, by—
 - (i) using barriers, signs, indicators and alarms; and
 - (ii) requiring stated clearance distances between plant and electrical conductors; and
 - (iii) isolating electrical conductors.

137 Fire detection and suppression in tunnels

A surface mine's safety and health management system must provide for automatically detecting combustion and suppressing fires in tunnels beneath stockpiles.

144 Lighting

A surface mine's safety and health management system must provide for the following—

- (a) adequate lighting, having regard to shadow, contrast and glare, in each area of the mine in which work is to be carried out so the work can be carried out safely;
- (b) emergency lighting to ensure that persons in a building or enclosure at the mine can safely exit if the normal lighting fails.

146 Trenches

- (1) A surface mine's safety and health management system must provide for protecting persons from risks from trenches at the mine.
- (2) The system must provide for the following—
 - (a) the use of barricades;
 - (b) ground support;
 - (c) water ingress and hazardous atmosphere in the trench.

370A Activity that is not an on-site activity—[Act, s 10](#)

(1) For [section 10\(2\)\(g\)](#) of the [Act](#), an activity performed by an electricity entity, in relation to its works, and for the purpose of supplying electricity, is not an on-site activity.

(2) In this section—

electricity entity has the meaning given by the [Electrical Safety Act 2002](#).

works, of an electricity entity, has the meaning given by the [Electrical Safety Act 2002](#), [section 25](#).